KEVIN LEVY

kelevy96@gmail.com | +352-621754511 | LinkedIn | GitHub| Website

PROFILE

Data Scientist with a PhD in Astrophysics and 4+ years of professional experience across academia and industry. Skilled in applying statistical and machine learning techniques to extract insights from complex, noisy data, with a strong emphasis on rigorous validation and bias control.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, Scikit-learn, PyTorch, Matplotlib), SQL, Shell Scripting
- **Data Analysis**: Data Cleaning & Visualization, Exploratory Data Analysis (EDA), Time Series Analysis
- **Statistics**: Maximum Likelihood Estimation (MLE), Bayesian Inference, MCMC Sampling, Hypothesis Testing
- **Machine Learning**: Linear/Logistic Regression, Gradient Boosting, K-Means, GMM, CNNs, LSTMs
- **Infrastructure & Computing**: Linux, High-Performance Computing (HPC), Docker, Git/GitHub
- **Cloud Platforms:** Google Cloud Platform (GCP; Vertex AI, BigQuery, Looker, Cloud Storage)

PROFESSIONAL EXPERIENCE

Doctoral Researcher, Astrophysics — University of Melbourne, Melbourne, Australia Jun 2022 – Jun 2026

- **Pipeline Development**: Led the development of two scientific end-to-end Python pipelines:
  1. Reconstructed and statistically characterized the Universe's matter distribution from a new South Pole Telescope dataset, validating its quality for high-precision cosmological analysis;
  2. Extracted unbiased galaxy-cluster mass estimates from millimeter-wave data, tested on simulations and used to forecast performance for future surveys. The pipeline was later applied to survey data, achieving a ~2× improvement in measurement precision.
- **Statistical Methods**: Applied maximum-likelihood estimation, jackknife resampling, and chi-square/PTE statistics for model fitting, covariance estimation, and detection significance.
- **Validation & Robustness**: Designed and ran a comprehensive suite of systematic tests to identify potential biases and ensure robust, reproducible measurements.
- **Collaboration & Communication**: Worked within an international team on shared, version-controlled codebases and presented results at international conferences.
- **Publications**: Levy et al. 2026, Levy et al. 2023

BI Developer — Delta Information Technologies, Mamer, Luxembourg Dec 2021 – May 2022

- **Data Engineering & Visualization**: Used SQL to build and maintain database views integrated with the company's ERP platform, developing dashboards and charts to track financial, buying, and selling KPIs across the client base.
- **Stakeholder Management**: Worked directly with clients on-site and via calls/Zoom to gather requirements, customizing dashboards to match each client's specific needs and preferences.

SELECTED INDEPENDENT PROJECT

Flood Detection from Satellite Imagery

- Built an end-to-end remote sensing pipeline for flood detection by training a PyTorch U-Net from Sentinel-1 SAR imagery, achieving 79% mean IoU (segmentation accuracy).
- Deployed the model as a live, publicly callable API on Google Cloud Run, using a custom containerized training job on Vertex AI for scalable training.

EDUCATION

- Ph.D. in Astrophysics — University of Melbourne, Melbourne, Australia Jun 2022 – Jun 2026
- M.Sc. in Astrophysics — University of Bonn, Bonn, Germany Oct 2019 – Oct 2021
- B.Sc. in Physics — University of Bonn, Bonn, Germany Oct 2015 – Oct 2019

CERTIFICATIONS & AWARDS

- Data Scientist in Python Path — Dataquest.io (2021)
- N.D. Goldsworthy Scholarship (2022) — Awarded on the basis of academic excellence

LANGUAGES

Luxembourgish (Native) · English (C2) · German (C2) · French (B2)